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A new deep clustering method with application to customer selection for demand response program

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ABSTRACT

Demand response (DR) is regarded as a promising solution to the problem of renewable energy integration, while it remains one of the key barriers for DR to target the right customers, i.e. the customers with high potential in peak reduction. Exploring the electricity consumption behaviors of customers can help DR program practitioners evaluate customers' peak reduction potential. Clustering has been the most common approach to describe customers' consumption behaviors in the literature. However, the existing clustering methods suffer from either unscalability or poor performance when handling a large number of daily load curves. To overcome these issues, a new deep learning-based clustering method is proposed in this paper, which integrates both intra-cluster compactness and inter-cluster separation into the objective function of clustering for consideration. Based on the clustering results, a new customer selection framework is developed to identify the potential good candidates for DR programs, which takes the stability of consumption behaviors and the characteristics at peak hours into account. Case studies on a London dataset demonstrate the superiority of the proposed clustering method over classical K-means and two state-of-the-art deep clustering methods in terms of clustering validity indexes. Furthermore, the simulations performed on an actual price-based pilot show that the customers selected by the proposed customer selection framework contribute to 59 % of total peak reduction with only accounting for 23 % of population.

1. Introduction

With energy crisis and climate change, developing renewable electricity has drawn attention from the whole world. The widely used renewable energy primarily include wind power, solar power and hydrogen energy, etc. Survey shows that, the share of electricity generation from renewable energy rose from 24.6 % in 2015 to 33 % in 2018 across the United Kingdom [1]. However, due to the uncertainty and randomness characteristics, the growing penetration of renewable energy brings great challenges to the balance between supply and demand, which is the foundation of power system operation [2]. Hence, demand response (DR) technologies, as a promising solution to the problem of renewable energy integration, develop rapidly all over the world [3].

DR programs aim to modify the consumption patterns of electricity by encouraging customers to reduce or shift their electricity demand at peak time in response to price signals or financial incentives [3]. Generally, the DR programs are divided into two major categories: incentive-based programs and price-based programs [4]. To be more specific, incentive-based programs, such as direct load control and demand bidding, provide fixed or time-varying incentives with customers for their demand changes during hours of system stress. Price-based programs directly send customers time-varying electricity price signals to encourage modification of their consumption patterns. For the price-based programs, there are various types of dynamic pricing schemes, including critical-peak pricing, real-time pricing and time-of-use tariffs [5].

One of the significant barriers for DR programs is to target the right customers, i.e. the customers with high potential in peak reduction, which is critical to increase the profits of utilities and achieve bill reduction for individual customers [6]. To evaluate the peak reduction potential of customers, their electricity consumption behaviors require to be explored. Traditionally, the primary methods to describe these behaviors are through monthly billing data or manually conducted surveys, which are hard to provide an accurate description of consumption behaviors due to the incompleteness and low resolution of these data [7]. Recently the vast deployment of smart meters has

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generated a huge amount of fine-grained electricity consumption data available to the utilities, opening up opportunities to describe customers' consumption behaviors in a more accurate way [8]. The knowledge derived from the study can help DR program practitioners develop an effective customer selection strategy to identify the potential good candidates.

Clustering has been the most common approach to describe customers' electricity consumption behaviors in literature [9]. Most existing studies have focused on using the typical load curves to represent customers' consumption behaviors [10]. The typical load curve of an individual customer is usually obtained by averaging the daily load curves (DLCs) within a specific period (e.g. a month or a year). Then, instead of handling the large number of DLCs, the description of electricity consumption behaviors can be achieved by using the limited number of typical load curves. Although this approach significantly compresses the data, it also eliminates the daily and intra-day variations within each customer, resulting in an unreal reflection of customers' behaviors [11]. Compared to the compressed data, the original DLCs that retain the temporal variations between different consumption behaviors, are more valuable to be studied.

However, the considerable amount of original DLCs has disabled quite a few clustering algorithms, including hierarchical clustering [12], affinity propagation clustering [13], spectral clustering [14] and K-Medoids [11]. These algorithms require the distance between any two samples in the dataset for implementation, and thus cannot scale well to the large datasets. To overcome the issue of scalability, some feasible alternatives including K-means [15], adaptive K-means [16], fuzzy C-means (FCM) [17] and K-shape [18], are applied to clustering the large sets of DLCs. However, the scalability of these algorithms is typically acquired at the expense of clustering quality. For example, most algorithms in K-means family depend on only the intra-cluster compactness while ignoring the inter-cluster separation, leading to poor performance [19].

As mentioned above, traditional clustering methods typically suffer from either unscalability or poor performance when handling the large datasets. Therefore, further progress has been made on deep learning-based clustering methods, namely deep clustering methods. Existing deep clustering methods can be classified into two categories: two-stage methods and integrated methods [20]. The two-stage methods first perform feature extraction by various deep neural networks and then cluster the extracted features by traditional clustering algorithms. For example, [21] uses a convolutional autoencoder to extract features from yearly load curves and then applies K-means for clustering. The integrated methods combine the feature extraction stage and the clustering stage into a singe end-to-end model. Deep embedding clustering (DEC) [22], as one of the representative integrated methods, has been used in [5,23,24] to cluster DLCs or yearly load curves, demonstrating the superior performance over the traditional clustering methods. Further, to preserve the local structure in data, an improved deep embedded clustering (IDEC) is proposed by [25], which shows better performance than DEC in image and text clustering tasks.

Except for learning clustering-friendly features, achieving a good balance between the intra-cluster compactness and inter-cluster separation is also crucial to a good clustering quality [26]. Compactness represents how similar the objects in the same cluster are while separation indicates how different the objects in the different clusters are. Only considering the compactness may lead to the clusters that are quite similar to each other, while only considering the separation may result in the clusters that are substantially affected by the outliers. The development of deep learning not only makes it possible to optimize feature extraction and clustering in a single model, but also makes it possible to optimize intra-cluster compactness and inter-cluster separation in a single model. However, few deep clustering algorithms integrate both of these two aspects into the objective function of clustering for consideration.

Based on the results of DLCs clustering, certain customer selection strategies can be developed. The stability of consumption behaviors is considered to be an important indicator of customers' peak reduction potential. By applying clustering methods, the quantification of behavior stability can be achieved [4,27,28,29]. However, how it relates to peak reduction potential still remains controversial. Some studies [4,27] assume that the customers who follow consistent load patterns are good candidates for price-based DR programs, while others [28,29] assume that the customers who change load patterns frequently are more suitable for price-based DR programs. Unfortunately, in these studies, only [27] has validated the assumption on an actual pilot program in Ireland. In [27], comparisons are made between the stable customers and the all customers for the consumption change at peak hours. The results show that the stable ones result in more peak reduction under time-of-use tariffs. Besides the behavior stability, the characteristic at peak hours is another important indicator of customers' peak reduction potential. [30] clusters the customers of a pilot program in Korea into various groups, and the groups that show consistent peak demand occurrence times with the power system achieve much more peak reduction than others. Furthermore, some studies have applied supervised learning techniques to customer selection for DR programs. [31] uses historical consumption data and random forest to predict whether a customer would contribute to high peak reduction under time-of-use tariffs. [3,32] use feature selection methods and association rule mining methods respectively to explore the drivers of customers' peak reduction potential, including socio-demographic factors and dwelling characteristics. Overall, compared to the clustering technique, the applications of supervised learning techniques are still relatively limited, which is probably because the sample size of pilot datasets is usually insufficient for supervised learning models.

Based on the aforementioned discussion, this paper proposes a new deep clustering method with its application to customer selection for DR programs. For a good clustering quality, both of intra-cluster compactness and inter-cluster separation are integrated into the objective function of clustering to seek a good balance between them. After obtaining the clustering results, a new customer selection framework is developed to identify the customers with high potential in peak reduction, which takes the behavior stability and the peak characteristics into account. Case studies demonstrate the superiority of the proposed deep clustering method and the effectiveness of the proposed customer selection framework. Therefore, the main contributions of this paper can be summarized as follows:

- A deep clustering method is proposed for a large number of DLCs to improve the clustering quality. Case studies on a London dataset demonstrate the superiority of the proposed method over classical K-means and two state-of-the-art deep clustering methods (DEC [5] and IDEC [25]) in terms of clustering validity indexes.
- A customer selection framework is developed to identify the potential good candidates for DR programs. Simulations performed on an actual price-based pilot show that the customers selected by the proposed framework contribute to 59 % of total peak reduction with only accounting for 23 % of population.

The rest of this paper is organized as follows. Section 2 gives the description of the pilot program and dataset used in this paper. Section 3 presents the details of the proposed deep clustering method. Section 4 presents the details of the proposed customer selection framework. Section 5 presents simulation studies and comparative results. Section 6 concludes the paper.

2. Pilot program and dataset

The electricity consumption data used in this paper is obtained from the Low Carbon London program in the United Kingdom [33]. The dataset includes the electricity consumption data of 5567 residential customers at 30 min interval from July in 2012 to December in 2013,

including the benchmark period (July 2012-Dec 2012) and the test period (Jan 2013-Dec 2013). Customers were assigned to a test or control group. During the benchmark period, all the customers followed their usual electricity consumption regime with a standard flat tariff. During the test period, customers of test group were asked to trial a time-of-use (ToU) tariff while the customers in the control group still followed their usual consumption regime. The details of the standard flat tariff and the intra-day averaged ToU tariff are presented in Fig. 1. It can be seen that the price rates of ToU tariff are much lower than those of standard flat tariff at most hours, while during 5:00–10:00p.m. the price rates of ToU tariff are significantly higher, which corresponds to the peak hours of local power system. After excluding the customers without intact electricity data, only 4627 customers remained, including 953 customers of test group and 3674 customers of control group.

3. Deep learning-based clustering

3.1. Preprocessing of data

In order to describe the patterns of electricity consumption, the impact caused by different amplitudes in DLCs needs to be removed through the normalization method before clustering. Each DLC is normalized by its total daily electricity consumption value as follows:

$$x_i(t) = \frac{h_i(t)}{\sum_{t=1}^{48} h_i(t)} \tag{1}$$

where $h_i(t)$ and $x_i(t)$ are the actual and normalized active power at time$t$. Similar to [10], x_i implies a daily electricity consumption pattern (DECP). In the following, the clustering is performed on DECPs.

3.2. Model architecture

Let $X = [x_1, \ldots, x_N]^T$ denote the DECPs that need to be clustered, where x_i denotes a DECP and N denotes the number of DECPs. Let K denote the number of clusters. As shown in Fig. 2, the model architecture of the proposed method consists of two major components: the assignment network and the centroid layer.

- Assignment network

It is composed of fully connected layers of neural networks and is used to learn the relationship between DECPs and clusters. Specifically, the assignment network takes x_i as input and estimates the probabilities of assigning it to each cluster. As the summation of the estimated probabilities for any x_i must be consistently 1, the softmax function [34] is employed as the activation function of the last layer to enforce the property of the probability distribution. Therefore, for the inputx_i, the output of assignment network is a vector of length$K$, expressed as$s_i = [s_{i1}, \ldots, s_{i,k}]$. $s_{i,k}$ denotes the probability of assigning x_i to cluster$k$, and is calculated as follows:

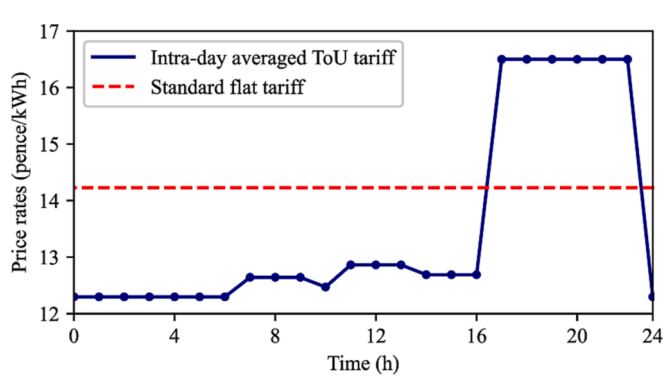


Fig. 1. Price rates of the standard flat tariff and the intra-day averaged ToU tariff.

$$s_{i,k} = \frac{e^{o_k(x_i)}}{\sum_{j=1}^{K} e^{o_j(x_i)}} \tag{2}$$

where $o_j(x_i)$ is the output value of the jth neuron in the last layer with the inputx_i. For each samplex_i, the cluster label y_i is obtained by finding the optimal index k^* that corresponds to the maximum value of all $s_{i,k}$ for $k = 1, \ldots, K$ (i.e. $y_i = arg\max_k s_{i,k}$). Therefore, all DECPs are clustered into K clusters via the assignment network.

- Centroid layer

It contains only one custom network layer, whose parameters are composed of the cluster centroids of K clusters, expressed as$[\mu_1, \ldots, \mu_K]^T$. After x_i is input, the centroid layer firstly calculates the distance between x_i and each centroid, and then calculates the distance between centroids themselves, as shown in Fig. 2. Therefore, the output of centroid layer contains two parts, which are expressed as $\|x_i - \mu_k\|$ and$\left\|\mu_k - \mu_j\right\|$, respectively, where $\|\cdot\|$ denotes l_2 norm and$k, j = 1, \ldots, K$.

3.3. Loss function

As a good clustering quality should achieve a good balance between the intra-cluster compactness and the inter-cluster separation, a compactness loss L_c and a separation loss L_s are defined to ensure the balance.

- Compactness loss

L_c is adapted from the loss function of K-means, which only takes the intra-cluster compactness into consideration. K-means loss can be expressed as follows:

$$\sum_{i=1}^{N} \sum_{k=1}^{K} z_{i,k} \|x_i - \mu_k\|^2 \tag{3}$$

where $z_{i,k}$ is a Boolean variable, taking value of one if x_i is assigned to cluster$k$, zero otherwise. By replacing $z_{i,k}$ with the probabilities estimated by the assignment network, L_c can be obtained as follows:

$$L_c = \sum_{i=1}^{N} \sum_{k=1}^{K} s_{i,k} \|x_i - \mu_k\|^2 \tag{4}$$

Similar to K-means, minimizing L_c reduces the intra-cluster sum of squared distance, which drives the samples in the same cluster to be more densely distributed.

- Separation loss

L_s is defined as the average penalties imposed on every-two centroids, which can be expressed as follows:

$$L_s = \frac{2}{K(K-1)} \sum_{1 \leqslant k < j}^{K} pen(\mu_j, \mu_k) \tag{5}$$

where $pen(\mu_j, \mu_k)$ denotes the penalty imposed on a pair of centroids μ_j and μ_k to avoid too small distance between them. [35] uses $-\left\|\mu_j - \mu_k\right\|$ as$pen(\mu_j, \mu_k)$, but this way may lead to poor performance as L_s can be minimized by increasing the distance between only a few centroids. To address this issue, this paper defines $pen(\mu_j, \mu_k)$ as follows:

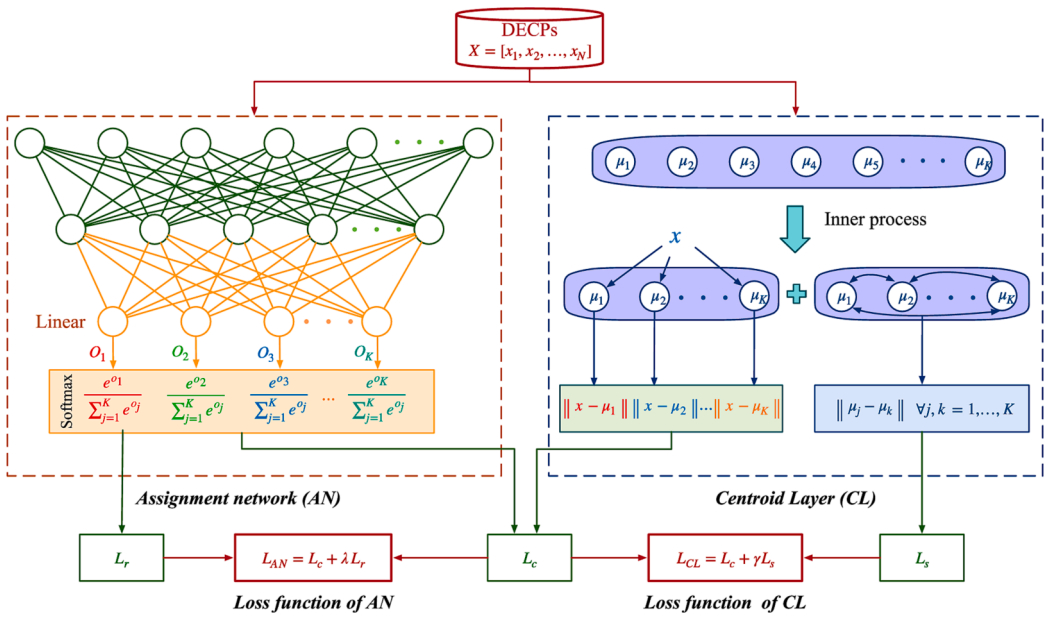


Fig. 2. The network structure of the proposed clustering method.

$$pen(\mu_j, \mu_k) = \begin{cases} \delta - \|\mu_j - \mu_k\| & \|\mu_j - \mu_k\| \langle \delta \\ 0 & \text{others} \end{cases} \tag{6}$$

where δ is a threshold that controls the minimum acceptable distance between a pair of centroids. In this way, only when every-two centroids keep a distance of at leastδ, can L_s reach the minimum. Subsequently, the distance between samples from different clusters is also increased.

- Loss function of centroid layer

As both L_s and L_c involve the cluster centroids, the loss function of the centroid layer can be obtained by combining them, as follows:

$$L_{CL} = L_c + \gamma L_s \tag{7}$$

where γ is a trade-off parameter used to balance L_c andL_s. An appropriate γ is crucial for the balance between intra-cluster compactness and inter-cluster separation.

- Loss function of assignment network

As L_c involves the probabilities estimated by the assignment network, the loss function of assignment network definitely includesL_c. To avoid the trivial solutions, where all the samples are assigned to only a few clusters, a regularization loss L_r is introduced to the assignment network. Before introducingL_r, the empirical distribution of cluster labels are firstly defined. The average probability of assigning each DECP to cluster k can be calculated as:

$$p_k = \frac{1}{N} \sum_{i=1}^{N} s_{i,k} \tag{8}$$

The values of p_k for $k = 1, \ldots, K$ constitute the empirical distribution of cluster labels P (i.e. $P = \{p_1, \ldots, p_K\}$). Then, the regularization loss L_r can be defined as the Kullback-Leibler (KL) divergence between P and the uniform distribution$U = \{\frac{1}{K}, \ldots, \frac{1}{K}\}$, as follows:

$$L_r = KL(U \| P) = \sum_{k=1}^{K} \frac{1}{K} \log \frac{\frac{1}{K}}{p_k} \tag{9}$$

L_r reaches the minimum only if all the samples are evenly distributed over K clusters, which helps avoid trivial solutions. By combining L_c andL_r, the loss function of assignment network can be obtained as follows:

$$L_{AN} = L_c + \lambda L_r \tag{10}$$

where λ is a trade-off parameter that controls the amount of regularization.

3.4. Parameter optimization

This paper uses mini-batch gradient descent and backpropagation to minimize L_{CL} and L_{AN} to find optimal cluster labels and cluster centroids. The optimization procedure is summarized in Algorithm 1, including three steps. In the first step, the classical K-means is performed to obtain K cluster centroids, which are used as the initial parameters of centroid layer. In the second step, pick a batch set X^S from$X$. Calculate the gradients of L_{CL} with respect to μ_k (i.e. $\frac{\partial L_{CL}}{\partial \mu_k}$) and the gradients of L_{AN} with respect to s_i (i.e. $\frac{\partial L_{AN}}{\partial s_i}$) based on$X^S$. Pass down $\frac{\partial L_{CL}}{\partial \mu_k}$ and $\frac{\partial L_{AN}}{\partial s_i}$ to update the cluster centroids and the assignment network parameters, respectively. In the third step, update the cluster label y_i for each x_i according to$y_i = argmax_k s_{i,k}$. Repeat the second step and the third step until less than ϵ% cluster labels are changed between every-two iterations or the optimization process has been run for $\overline{iter}$ iterations.

Algorithm 1 Proposed deep clustering algorithm

Input: DECP pool:$X = [x_1, \ldots, x_N]^T$; Number of clusters:K; Maximum number of iterations:$\overline{iter}$; Convergence criterion:ϵ%; Cluster labels update interval:I_U
Output: Cluster labels:Y
Initialize cluster centroids $[\mu_1, \ldots, \mu_K]^T$ via classical K-means.
$Y = Kmeans(X,K)$.
for $iter = 1$: $\overline{iter}$ **do**
if $Modulo(iter, I_U) == 0$ **then**
$Y^0 = Y$
Update cluster labels.
for $i = 1$: N **do**
$y_i = argmax_k s_{i,k}$
end for
/*less than ϵ% cluster labels are changed*/
if $\frac{\#(Y \neq Y^0)}{N} < \epsilon\%$ **then**
Stop
end if
end if

(continued on next page)

(*continued*)

Algorithm 1 Proposed deep clustering algorithm

14: Pick a batch set$X^S \subset X$.

15: Calculate and pass down the gradients $\frac{\partial L_{CL}}{\partial \mu_k}$ and $\frac{\partial L_{AN}}{\partial s_i}$ based on X^S to update the cluster centroids $[\mu_1, \ldots, \mu_K]^T$ and the parameters of assignment network, respectively.

16: **end for**

4. Application to DR programs

4.1. Targeting scenario

Consider a targeting scenario which occurs after the benchmark period ends but before the test period begins, where the DR program practitioners intend to target the customers with high potential in peak reduction. The targeting scenario occurrence time is shown in Fig. 3. It can be seen that when the targeting scenario occurs, only the benchmark period data are available to practitioners while the test period data are unavailable. Hence, the benchmark period data of the test group will be used to perform customer selection; and the validity of selection results can be evaluated by the actual peak reduction achieved by the selected customers during the test period.

4.2. Customer selection framework

For the abovementioned targeting scenario, a customer selection framework is proposed to identify the customers with high potential in peak reduction. Fig. 4 shows the schematic of the proposed framework, which consists of the following four steps:

- DECPs clustering

The proposed deep clustering method is used to partition the large number of DECPs into different clusters. After clustering, each cluster represents a unique electricity consumption pattern and each customer is represented by a percentage distribution of DECPs over clusters, which will be further used to measure the behaviour stability.

- Stability measuring

As mentioned in [4,27], the customers who follow consistent DECPs are regarded as stable customers who are suitable for DR programs. On the contrary, the customers showing high variability in DECPs are less likely to achieve persistent peak reduction as high variability indicates the presence of uncertainty factors that may lead to the unpredictable consumption behaviours. By applying the clustering method, the DECPs

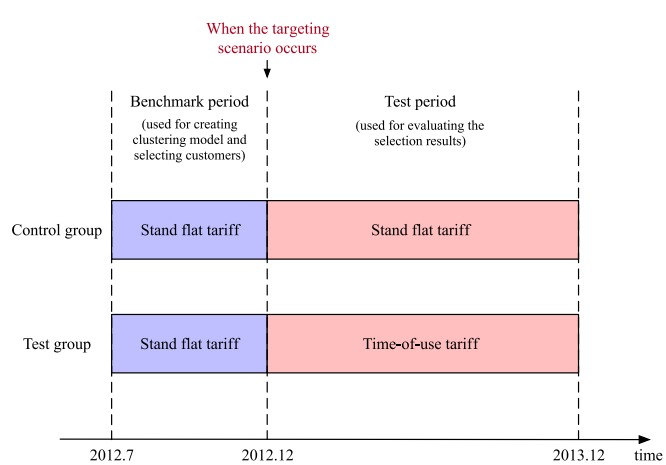


Fig. 3. The targeting scenario occurrence time.

of each customer are assigned to different clusters based on his/her consumption habits. The higher variability means that the DECPs are more randomly distributed over clusters. To quantify such variability, an information theory metric Shannon entropy [36] is employed, which is calculated by:

$$H = -\sum_{k=1}^{K} p(k)\log p(k) \tag{11}$$

where K is the cluster number, and $p(k)$ is the percentage of a customer's DECPs assigned to cluster$k$. The customers with smaller entropy are considered as the stable customers. They are more likely to achieve persistent peak reduction if they are selected for DR programs.

- Contribution evaluation

The more the customers contribute to the peak demand of power system, the higher potential they have in peak reduction. To provide an index to quantify such contribution, an intuitive way is to use the electricity consumption at peak hours, which usually refer to the evening time for residential customers. However, this method cannot offer a fair basis for comparing customers. For example, Fig. 5 shows the DLCs of two customers. In spite of less electricity consumed by customer B, he/she clearly has higher potential in peak reduction as most of his/her electricity is consumed at peak hours, while customer A consumes most of his/her electricity at non-peak hours. Compared to the consumption itself, it's fairer to use the consumption percentage at peak hours to compare customers. Therefore, a new index called contribution factor (CF), is defined as the ratio of peak consumption percentage of individual customers to that of power system, which can be calculated by:

$$CF_i = \frac{1}{n_{bp}} \sum_{d \in BP} \frac{f_{i,d}}{f_{ps,d}} \tag{12}$$

where CF_i denotes the contribution factor for customer$i$, BP denotes the benchmark period and n_{bp} denotes the number of days in the benchmark period. $f_{i,d}$ and $f_{ps,d}$ denote the peak consumption percentage at day d for the customer i and the power system, respectively. The customers with CF > 1 have electricity consumption more concentrated on peak hours than the average level, and consequently they have higher potential to reduce peak demand.

- Customer segmentation

By finding a proper threshold for each of entropy and contribution factor, the customers can be segmented into four categories. In each dimension, the customers are divided into two groups, one group with higher interest from practitioners and the other one with lower interest. In this way, customers who have stable consumption behaviours with high consumption percentage at peak hours can be selected as potential good candidates for DR programs.

- Evaluation of customer selection results

To evaluate the validity of customer selection results, the actual peak reduction achieved by the selected customers during the test period requires to be quantified. The percentage change in peak electricity consumption for customer i during the test period spanned by the ToU tariffs against the benchmark period is given by

$$\widetilde{R}_i = 100\% \times \frac{PU_i^{pre} - PU_i^{post}}{PU_i^{pre}} \tag{13}$$

where PU_i^{pre} denotes the average peak consumption of customer i during the benchmark period and PU_i^{post} is the counterpart for the test period.

However, during the test period, not only the ToU tariffs but also other external factors such as climate change, regional economy devel-

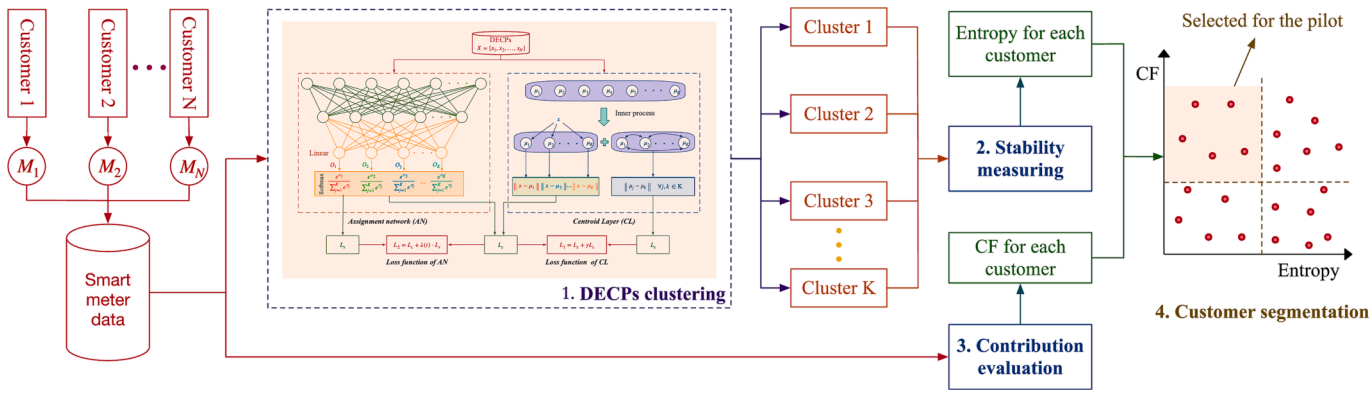
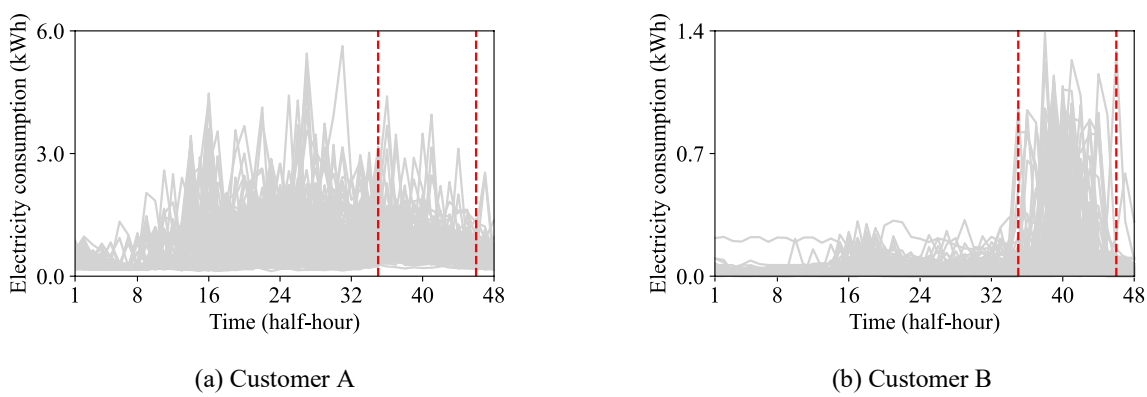


Fig. 4. The schematic of the proposed customer selection framework.



(a) Customer A (b) Customer B

Fig. 5. Daily load curves of two customers.

opment [37,38], may influence the peak consumption. To get an accurate evaluation of ToU tariff effects, we also calculate the percentage change between the test period and benchmark period for the control group. The percentage change of the control group is not affected by ToU tariffs and only reflects the effects caused by external factors, which is called common trend [3]. Considering that the control and test groups are both approximately representative of London [39], it's reasonable to assume that they share the same common trend during the test period. Therefore, by comparing the percentage change between the test and control groups, the variations caused by external factors can be largely excluded [31]. Accordingly, the percentage change caused by the ToU tariffs for customer i in the test group is given by

$$R_i = \widetilde{R}_i - \widetilde{R}_c \tag{14}$$

where $\widetilde{R}_c$ is given by

$$\widetilde{R}_c = \frac{1}{n_c}\sum_{i \in C}\widetilde{R}_i \tag{15}$$

where n_c denotes the number of customers in the control group. $\widetilde{R}_c$ denotes the average percentage change for customers of control group (C), who do not trial the ToU tariffs. After the percentage change R_i is obtained, the actual peak demand reduction caused by ToU tariffs for customer i can be calculated as follows:

$$PDR_i = PU_i^{pre} \cdot R_i \tag{16}$$

Therefore, the total PDR_i for the selected customers can be used to evaluate the validity of customer selection results.

5. Case study

5.1. Experimental setup

Two classical traditional clustering methods, K-means and FCM, and two state-of-the-art deep clustering methods, DEC and IDEC are chosen to compare with the proposed deep clustering method. The used DEC and IDEC are from the publicly available codes released by their authors. Determining hyperparameters by cross-validation is not an option for unsupervised clustering [22]. Thus, the hyperparameters are determined based on CVIs on the whole dataset. Specifically, the mean of standardized DBI and WCBCR is used to determine the best hyperparameters for three deep clustering methods. The hyperparameters of the proposed method are listed in Table 1. K-means and FCM are run 20 times with different initialization and the best results are recorded. All the experiments are conducted in Python3.6 environment with the main packages of scikit-learn, Keras and TensorFlow on an Intel-i7 personal computer with 12 GB RAM.

Table 1
Hyperparameters of the proposed deep clustering method.

Parameter description	Value
Number of hidden layers	2
Number of neurons	256–128
Batch size	256
Convergence criterion	0.1 %
Optimizer	Adam
Learning rate	0.001
Maximum iterations	4e4
Cluster labels update interval	500
λ in (10)	0.1
γ in (7)	1

5.2. Evaluation metrics

Four clustering validity indexes (CVIs) applied in many works [26,40] are adopted to evaluate the clustering performance:

- Compactness (Comp): Comp is defined as the average distance between samples and their corresponding centroids, which can be calculated by:

$$\mathrm{Comp} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|C_k|}\sum_{x\in C_k} d(x,\mu_k)^2 \tag{17}$$

where K denotes the number of clusters, μ_k denotes the centroid of clusterC_k, $|C_k|$ denotes the number of samples in cluster C_k and $d(x,y)$ denotes the distance between samples x and$y$.

- Separation (Sep): Sep is defined as the average distance between different centroids, which can be calculated by:

$$\mathrm{Sep} = \frac{2}{K(K-1)}\sum_{1\leqslant q<p}^{K} d\left(\mu_p,\mu_q\right)^2 \tag{18}$$

- Davies-Bouldin index (DBI): DBI is defined as the ratio of the within cluster scatter and the separation between clusters, which can be calculated by:

$$\mathrm{DBI} = \frac{1}{K}\sum_{k=1}^{K}\max_{j\neq k}\left\{\frac{\frac{1}{n_k}\sum_{x\in C_k} d(x,\mu_i) + \frac{1}{n_j}\sum_{x\in C_j} d\left(x,\mu_j\right)}{d\left(\mu_k,\mu_j\right)}\right\} \tag{19}$$

where n_k denotes the number of samples in cluster$k$.

- Ratio of within cluster sum of squares to between cluster variation (WCBCR): WCBCR is defined as the ratio of the sum of the squared distance between each sample and its corresponding centroid to the sum of the squared distance between centroids, which can be calculated by:

$$\mathrm{WCBCR} = \frac{\sum_{k=1}^{K}\sum_{x\in C_k} d^2(x,\mu_k)}{\sum_{k=1}^{K}\sum_{1\leqslant j\leqslant k} d^2\left(\mu_j,\mu_k\right)} \tag{20}$$

It should be noted that Comp and Sep are adopted to provide an evaluation of the intra-cluster compactness and the inter-cluster separation, respectively. DBI and WCBCR are adopted to provide a comprehensive evaluation of these two aspects. In these CVIs, larger values of Sep indicate better inter-cluster separation, while smaller values of the other CVIs stand for better effects.

5.3. Clustering performance evaluation

- Comparison with the state-of-the-art

A wide range of numbers of clusters$K$ = [2,4,6,8,10,20,40,60,80,100] are chosen to show the effectiveness of the proposed clustering method. Table 2 shows the values of four CVIs for each clustering method. FCM and K-means produce the best results in terms of compactness but relatively poor results in terms of separation, which is expected because their objective functions only take the intra-cluster compactness into consideration, while ignoring the inter-cluster separation. The compactness produced by the proposed method is between DEC and IDEC, while the separation produced by the proposed method outperforms the other competing algorithms, which indicates the clusters obtained by the proposed method are more dispersive. In terms of the comprehensive CVIs, i.e. DBI and WCBCR, the proposed method shows the best performance, irrespective of the variations in cluster numbers. Specifically, compared to K-means, DEC and IDEC, the proposed method improves the DBI by around 16.1 %, 11.0 % and 16.5

Table 2
CVIs values for different clustering methods with a wide range of cluster numbers.

Method	K	Comp(↓)	Sep(↑)	DBI(↓)	WCBCR(↓)	K	Comp(↓)	Sep(↑)	DBI(↓)	WCBCR(↓)
FCM	2	1.510	0.393	3.509	5.15E + 06	20	**1.148**	0.608	4.813	1.84E + 04
K-means		**1.506**	0.430	3.366	4.69E + 06		1.454	1.978	2.349	3.82E + 03
DEC		1.710	0.623	2.827	3.42E + 06		1.563	2.394	2.337	3.16E + 03
IDEC		1.810	0.731	2.665	3.51E + 06		1.654	2.350	2.391	3.65E + 03
Proposed		1.749	**0.945**	**2.488**	**2.19E + 06**		1.792	**3.262**	**2.045**	**2.42E + 03**
FCM	4	1.467	0.377	4.181	8.68E + 05	40	**1.046**	0.654	4.618	4.58E + 03
K-means		**1.407**	0.587	3.100	5.31E + 05		1.334	2.615	2.179	6.16E + 02
DEC		1.893	0.922	2.765	2.96E + 05		1.425	3.019	2.150	5.37E + 02
IDEC		2.172	0.884	2.891	4.89E + 05		1.577	2.490	2.380	6.88E + 02
Proposed		2.002	**1.679**	**2.382**	**1.95E + 05**		1.572	**3.804**	**1.935**	**4.60E + 02**
FCM	6	**1.324**	0.417	4.575	3.17E + 05	60	**0.999**	0.712	4.805	2.38E + 03
K-means		1.516	1.829	2.636	6.42E + 04		1.289	3.096	2.244	2.13E + 02
DEC		1.687	2.066	2.418	6.02E + 04		1.441	3.482	2.181	2.09E + 02
IDEC		2.175	1.567	2.847	7.07E + 04		1.630	2.914	2.320	2.56E + 02
Proposed		1.960	**2.142**	**2.160**	**5.79E + 04**		1.527	**4.055**	**1.965**	**1.83E + 02**
FCM	8	**1.380**	0.535	4.121	1.27E + 05	80	**0.889**	1.048	4.184	1.02E + 03
K-means		1.541	1.755	2.572	3.42E + 04		1.267	3.523	2.225	9.99E + 01
DEC		1.624	1.938	2.390	3.11E + 04		1.486	3.732	2.163	9.51E + 01
IDEC		1.716	1.772	2.710	4.16E + 04		1.530	3.215	2.412	1.27E + 02
Proposed		2.025	**2.255**	**2.257**	**2.80E + 04**		1.460	**3.875**	**2.005**	**8.90E + 01**
FCM	10	**1.237**	0.585	4.210	7.13E + 04	100	**0.991**	0.863	4.634	1.07E + 03
K-means		1.503	1.723	2.520	2.09E + 04		1.228	4.371	2.209	5.00E + 01
DEC		1.601	2.149	2.362	1.85E + 04		1.331	4.529	2.153	4.87E + 01
IDEC		1.722	2.297	2.503	1.97E + 04		1.401	4.048	2.169	6.20E + 01
Proposed		2.200	**3.333**	**1.887**	**1.17E + 04**		1.417	**4.804**	**1.978**	**4.75E + 01**

The ↑ (or ↓) represents whether larger (or smaller) values indicate better effects.

% on average, respectively and improves the WCBCR by around 28.0 %, 17.8 % and 33.8 % on average, respectively. The superior performance of the proposed method demonstrates that it's effective to integrate both intra-cluster compactness and inter-cluster separation into the objective function for a good clustering quality, which is consistent of the finding in [26].

Apart from clustering quality, the computational efficiency is also important for evaluating the performance of clustering algorithms. Here, we compare the runtime of the proposed algorithm with that of other competing algorithms. The runtime is the average of runtime of clustering with$K$ = [2,4,6,8,10,20,40,60,80,100]. As K-means is substantially affected by initial centroids, it's typically run a large number of times with new initial centroids generated randomly at each round [9]. 20 rounds are performed here. The comparison results are shown in Table 3. It can be seen that K-means achieves the highest computational efficiency, while DEC and IDEC show the worst performance due to their high complexity in pretraining stage. The computational efficiency of the proposed method is between K-means and two deep clustering methods. Considering the superior clustering quality, the runtime of the proposed method might be acceptable.

- Sensitivity analysis

The proposed deep clustering method seeks a good balance between intra-cluster compactness and inter-cluster separation by finding an appropriate γ in (7). Therefore, it's interesting to investigate the effects of γ on the performance of the proposed method. Here, use $K = 30$ as an example and a wide range of values are selected forγ. A larger value of γ indicates that more weights are assigned to the inter-cluster separation against the intra-cluster compactness. The CVIs values for different values of γ are shown in Fig. 6. It can be seen that the separation increases significantly from $\gamma = 10^{-1}$ to$\gamma = 10^{0}$, while it changes slightly at other points. Different from the separation, the compactness remains a steady growth with the increase ofγ. In terms of DBI and WCBCR, they both reach the global minimum in$\gamma = 10^{0}$. Therefore, a good balance between intra-cluster compactness and inter-cluster separation is considered to be achieved when$\gamma = 10^{0}$.

Furthermore, λ in (10) controls the amount of regularization in the loss function of assignment network. A larger value of λ indicates that more weights are assigned to the regularization loss against the compactness loss. Here, we investigate the effects of λ on the performance of the proposed method, as shown in Table 4. It can be seen that if the values of λ are too large, such as$\lambda = 10^{1}, 10^{2}$, the clustering quality might be spoiled because the regularization loss tends to push all the samples to be evenly distributed over clusters regardless of intra-cluster compactness. On the other hand, if the values of λ are too small, such as$\lambda = 10^{-2}, 10^{-3}$, the trivial solutions might happen where all the samples are assigned to only a few clusters. Therefore, we choose $\lambda = 10^{-1}$ for (10) to obtain high clustering quality and avoid trivial solutions.

- Determination of cluster number

To find a suitable cluster number, both clustering quality and computation complexity require to be considered [26]. Clustering with small cluster number cannot discriminate different load patterns but excessive subdivision of clusters is also inappropriate for practical application. The values of DBI and WCBCR with$K$ = [2,4,6,8,10,20,40,60,80,100] are shown in Fig. 7. It can be seen that WCBCR presents a steady decline with the increase of cluster number, while DBI achieves the lowest value when the cluster number is 10. Although WCBCR can further decrease with more than 10 clusters, the computation complexity increases at the same time. Considering the balance between clustering quality and complexity, we finally set the number of clusters to be 10, which will be used for visualization and entropy calculation.

- Visualization of clustering results

To show the effectiveness of the proposed method in a vivid way, the cluster centroids obtained by the proposed method are shown along with those obtained by K-means. Here, as $K = 10$ obtains the lowest value of DBI, its corresponding centroids are used for visualization, as shown in Fig. 8. It can be seen that the centroids obtained by the proposed method reflect more distinct electricity consumption patterns as they have more prominent peaks and troughs. Additionally, it's found that the peak features can effectively characterize the differences between consumption patterns, which is consistent with the findings of [10,16]. Based on the time of peaks, ten clusters can be segmented into four categories, as

Table 3
Average runtime and percentual runtime with respect to K-means.

	Average runtime (s)	Percentual runtime with respect to K-means
K-means (20 rounds)	151.5	1
DEC	1074.0	7.09
IDEC	1188.6	7.85
Proposed	420.4	2.77

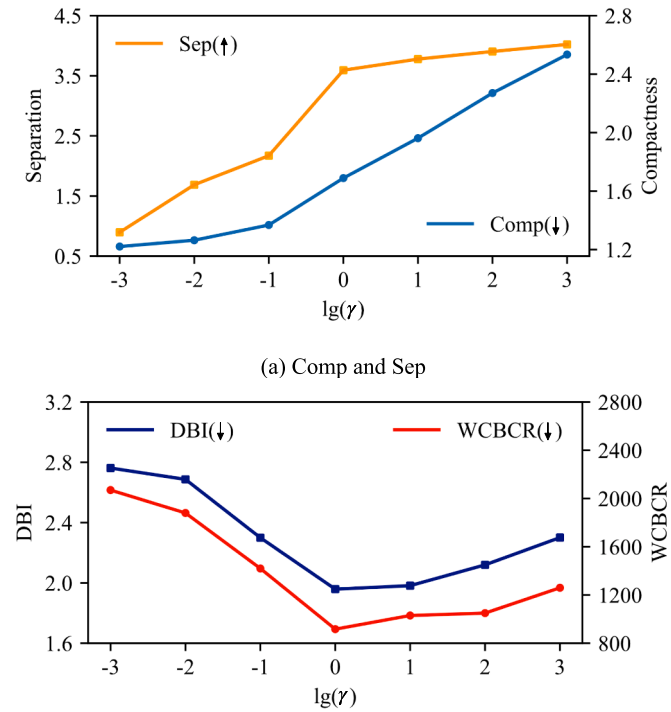
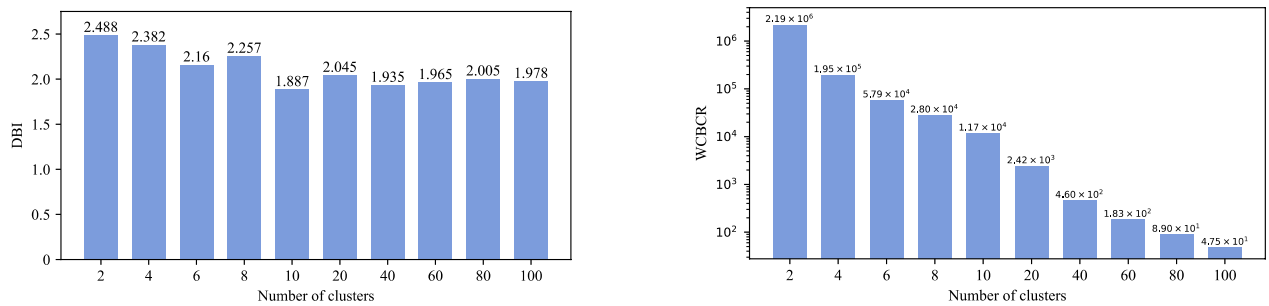


Fig. 6. CVIs values for different values of γ in (7). The ↑ (or ↓) represents whether larger (or smaller) values indicate better effects.

Table 4
Model performance for different values of λ in (10).

lg(λ)	2	1	0	−1	−2	−3
DBI(↓)	2.95	2.68	2.24	1.96	–	–
WCBCR(↓)	26.7	20.04	13.76	9.17	–	–

'↓' represents that smaller values indicate better effects.
'-' represents the occurrence of trivial solutions.



(a) DBI (b) WCBCR

Fig. 7. DBI and WCBCR values with a wide range of cluster numbers.

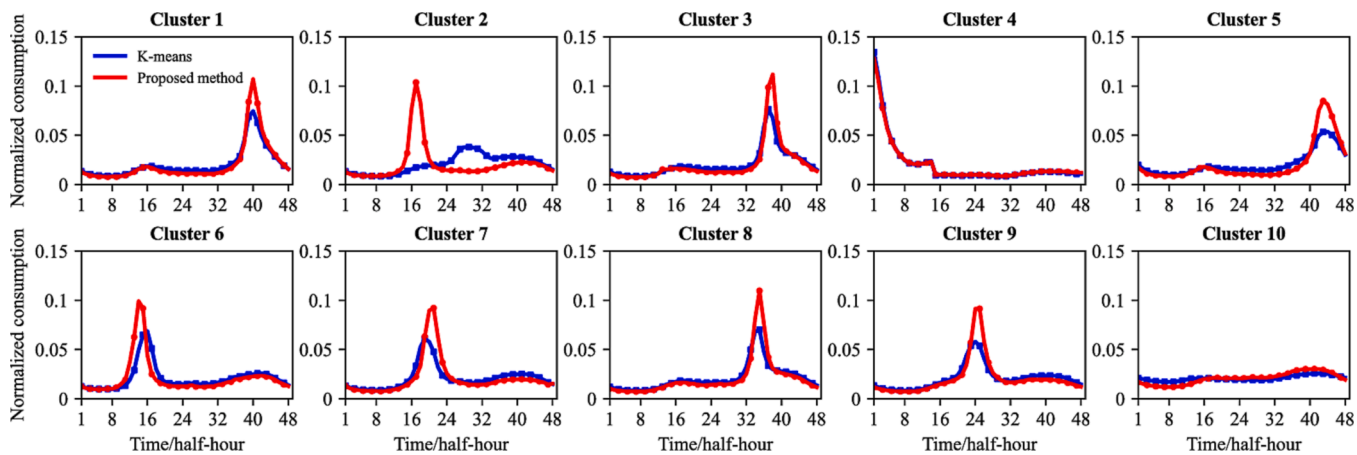


Fig. 8. The centroids of ten clusters obtained by K-means and the proposed clustering method, respectively.

listed in Table 5. It can be seen that Evening peak explains the most DECPs, followed by Morning peak and Daytime peak, while Night peak explains the least DECPs.

5.4. Applications to DR programs

Applying the proposed customer selection approach, the entropy and contribution factor can be calculated for each customer in the test group. Fig. 9 shows a scatter plot where each point represents a customer's entropy and contribution factor. According to this plot, customers can be divided into four categories by selecting a classification threshold for each dimension. Specifically, the median for entropy and "1″ for contribution factor are selected as the classification thresholds. For the convenience of further analysis, four categories are named as Group 1 to Group 4 (G1 to G4), as shown in Fig. 9. G1 contains customers with variable behaviors and high peak consumption percentage. Customers in G2 have similarly high peak consumption percentage as G1 but they show much more stable behaviors. G3 and G4 have totally opposite characteristics as G1 and G2. According to the proposed approach, customers in G2 have relatively high potential in peak reduction and should be selected as potential good candidates for the DR programs.

Table 5
Characteristics of four electricity consumption pattern categories.

Categories	Morning peak	Daytime peak	Evening peak	Night peak
Peak time	4:00–10:00	10:00–16:00	16:00–22:00	22:00–4:00
Cluster	2, 6	7, 9	1, 3, 5, 8, 10	4
Percent of DECPs	17.60 %	13.70 %	64.20 %	4.50 %

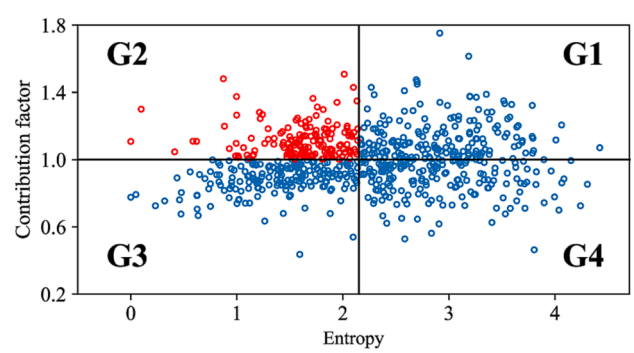


Fig. 9. The distribution of the classification results of the four customer categories. The points in red represent the customers selected for the pilot.

To evaluate the validity of customer selection results, the actual peak reduction is calculated for each customer in the test group using (13)-(16). Fig. 10 shows the frequency histogram of actual peak reduction for each customer category. It can be seen that each category shows positive peak reduction on average, but there are quite a few customers in G1, G3 and G4, who show increases in peak consumption under the ToU tariffs. Compared to the other three categories, G2 achieves significantly more peak reduction. Further, Table 6 presents the proportion of both population and total peak reduction for each customer category. It can be seen that G1 and G3 respectively account for around 26 % of population and contribute to around 17 % of total peak reduction. G4 accounts for 23.61 % of population while only 5.72 % of total peak reduction is from G4. In contrast to G4, G2 contains the similar number of customers as G4 but contributes to 59.32 % of total peak reduction. Additionally, the all

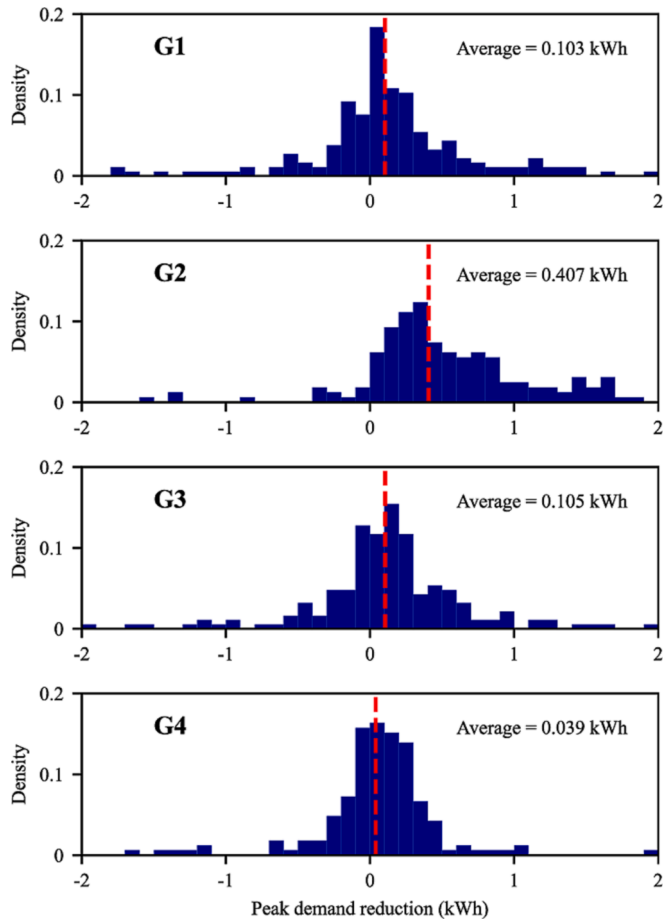


Fig. 10. The frequency histogram of peak demand reduction for each customer category. The dotted lines in red represent the average values.

Table 6
The number of customers and peak reduction in each customer category.

	G1	G2	G3	G4	All
Number of customers	252	220	256	225	953
Proportion of population	26.44 %	23.08 %	26.86 %	23.61 %	100 %
Average peak reduction (kWh)	0.103	0.407	0.105	0.039	0.159
Total peak reduction (kWh)	19.10	65.96	19.78	6.36	111.20
Proportion of total peak reduction	17.17 %	59.32 %	17.79 %	5.72 %	100 %

customers reduce the peak demand by 0.159 kWh per customer, which can be regarded as the results of random selection, while the customers selected by the proposed approach (i.e. customers in G2) reduce the peak demand by 0.407 kWh per customer, showing an improvement of 156 % compared to random selection.

6. Conclusion

This paper proposes a new deep clustering method for the large number of DECPs to improve the clustering quality, which integrates both intra-cluster compactness and inter-cluster separation into the objective function of clustering for consideration. After obtaining the clustering results, a customer selection framework is developed to identify the potential good candidates for DR programs, which takes the behaviour stability and peak characteristics into account. Case studies using the load data from London demonstrate the superiority and effectiveness of the proposed deep clustering method. Also, the simulations performed on an actual price-based pilot show that the customers selected by the proposed customer selection framework contribute to 59 % of total peak reduction with only accounting for 23 % of population.

In this paper, the proposed customer selection framework can only provide deterministic selection results, where the uncertainties about selection results are ignored. Considering the fact that customers' consumption behaviours have inherent randomness; and the selection results will be used to guide the decision-making for DR programs, it is crucial to quantify the uncertainties related to the selection results. In addition, the simulation results confirm that customers' capacity in peak reduction is highly correlated to their behaviour stability. However, as far as the authors know, there is no existing literature investigating how customers' behaviour stability relates to household characteristics, such as dwelling characteristics and socio-demographic factors. Therefore, the future work will also focus on exploring the relationship between customers' behaviour stability and household characteristics.

CRediT authorship contribution statement

Jiang-Wen Xiao: Supervision, Writing – review & editing, Funding acquisition. **Yutao Xie:** Writing – original draft, Methodology. **Hongliang Fang:** Supervision, Conceptualization, Writing – review & editing. **Yan-Wu Wang:** Conceptualization, Data curation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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